Supplementary material for:
A hybrid, bottom-up, structurally accurate, Gō-like coarse-grained protein model

Tanmoy Sanyal,[1, a)] Jeetain Mittal,[2] and M. Scott Shell[3, b)]
[1)] *Department of Chemical Engineering, University of California Santa Barbara, Santa Barbara, CA 93106, USA*
[2)] *Department of Chemical Engineering, Lehigh University, Bethlehem, PA 18015, USA*
[3)] *Department of Chemical Engineering, University of California Santa Barbara, Santa Barbara, CA 93106, USA.*

(Dated: 17 June 2019)

I. DEVELOPMENT OF AN EXTENDED-ENSEMBLE RELATIVE ENTROPY COARSE-GRAINING ALGORITHM

The extended-ensemble coarse-graining method, developed by Mullinax and Noid seeks to parameterize a bottom-up coarse-grained (CG) model simultaneously from an *ensemble* of all-atom (AA) reference simulations. The ensemble of AA references may constitute the same system at different thermodynamic state points (temperature, density, etc.) or chemically different systems under the same conditions or a combination of both.[1,2]. In this paper, we combine AA references of polyleucine (exclusively α helical) and polyvaline (β-hairpin) to develop a hybrid CG protein backbone that encodes both α and β character. To this end, we formulate the extended ensemble method in terms of relative entropy minimization between the CG model and the ensemble of AA systems.

Consider N independent AA reference simulations, such that simulation k has n_k number of statistically-independent configurational snapshots and a microstate probability distribution given by P^k_{AA}. While parameterizing a CG model from any one of these at a time (say the kth AA system), the likelihood of the model given the AA configurations in the kth system is simply: $\prod_i P_{\mathrm{CG}}(\mathbf{R}_i \mid \boldsymbol{\lambda})$. Here $\mathbf{R}_i$ is the set of atomistic degrees of freedom from snapshot i projected or mapped on to the reduced set of CG degrees of freedom, and $\{\boldsymbol{\lambda}\}$ is the set of parameters describing the CG forcefield. Since the N simulations are independent, the likelihood for the model given the *extended-ensemble* of all N references is:

$$L = \prod_{k=1}^{N}\prod_{i=1}^{n_k} P_{\mathrm{CG}}(\mathbf{R}_i \mid \boldsymbol{\lambda})$$

This likelihood can be considered a measure of goodness of the CG model, i.e. the optimal CG model that retains maximum information *simultaneously from all N references*, is the one that maximizes the likelihood. In practice, we maximize the log-likelihood to obtain:

$$\arg\max_{\boldsymbol{\lambda}} \log L = \arg\max_{\boldsymbol{\lambda}} \sum_{k=1}^{N}\sum_{i=1}^{n_k} \log P_{\mathrm{CG}}(\mathbf{R}_i \mid \boldsymbol{\lambda})$$

The inner summation can be manipulated as:

$$\begin{aligned}\sum_i \log P_{\mathrm{CG}}(\mathbf{R}_i \mid \boldsymbol{\lambda}) &= n_k\left[\frac{1}{n_k}\sum_i \log P_{\mathrm{CG}}(\mathbf{R}_i \mid \boldsymbol{\lambda})\right] \\ &\approx n_k\left\langle \log P_{\mathrm{CG}}(\mathbf{R} \mid \boldsymbol{\lambda})\right\rangle_{\mathrm{AA},k} \\ &= -n_k\left\langle \log \frac{P^k_{\mathrm{AA}}(\mathbf{R})}{P_{\mathrm{CG}}(\mathbf{R} \mid \boldsymbol{\lambda})}\right\rangle_{\mathrm{AA},k} + n_k\left\langle \log P^k_{\mathrm{AA}}(\mathbf{R})\right\rangle_{\mathrm{AA},k}\end{aligned}$$

Here in the second step, we take the thermodynamic limit $n_k \to \infty$, i.e., we have a large number of snapshots from all the AA trajectories in the extended ensemble, which allows us to replace the sum with an expectation over the (atomistic) microstate probability distribution of system k, denoted by $\langle\rangle_{\mathrm{AA},k}$. The next step involves multiplication and division by $P^k_{\mathrm{AA}}(\mathbf{R})$. Thus, the sum splits into two parts, the first of which can be identified as the negative of the relative entropy ($S^{(k)}_{\mathrm{rel}}$) between the CG model and AA reference k, while the second term is the Gibbs entropy of reference k and does not depend on the CG model parameters. This results in,

$$\begin{aligned}\arg\max_{\boldsymbol{\lambda}} \log L &= \sum_{k=1}^{N} \arg\min_{\boldsymbol{\lambda}} \left\langle \log \frac{P^k_{\mathrm{AA}}(\mathbf{R})}{P_{\mathrm{CG}}(\mathbf{R} \mid \boldsymbol{\lambda})}\right\rangle_{\mathrm{AA},k} \\ &= \arg\min_{\boldsymbol{\lambda}} \sum_{k=1}^{N} S^{(k)}_{\mathrm{rel}}\end{aligned}$$

Thus, simultaneous relative entropy minimization from multiple reference simulations guarantees maximal likelihood of the CG model being simultaneously representative of all AA members in the extended ensemble, irrespective of thermodynamic state or chemical dissimilarities between the references.

II. SELECTING A PSEUDO-SIDE CHAIN FOR GLYCINE

Glycine has two equivalent hydrogens which make it achiral. The Gō models presented in this work assumes one of

[a)] Also at: Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA 94158, USA.
[b)] Electronic mail: Author to whom correspondence should be addressed. Electronic mail: shell@engineering.ucsb.edu

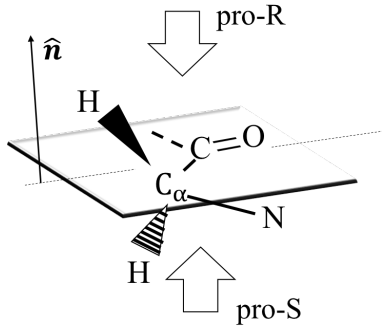

FIG. 1. The geometry of atomistic glycine in the reference plane of the sp^2 carbonyl carbon. Glycine has two equivalent hydrogens pointing along the outward (marked $\hat{\mathbf{n}}$) and inward normals to this plane. The Cahn-Ingold-Prelog stereochemical rules reveal the prochiral-S hydrogen as the one whose projection along the inward normal has a positive component.

these hydrogens as a pseudo sidechain for glycine for use in determining native contact pairs where glycine is a contacting partner. The location of the pseudo sidechain is determined as the hydrogen whose substitution with a higher group converts the stereochemistry to the L form which is the typical enantiomer for natural amino acids. Such a hydrogen is known as a prochiral-S hydrogen (S equivalent to L here) and can be determined from the Cahn Ingold Prelog rules[3] as the one that points away from the outward normal to the peptide bond plane.

III. THE $\mathrm{LEU}_{15}+\mathrm{VAL}_{15}$ CG BACKBONE AND Gō-LIKE SIDECHAIN FORCEFIELDS

The Gō-like native interactions for the various backbones have been presented in the manuscript. Here, we show full forcefield for the relative entropy optimized CG polypeptide parameterized from the joint ensemble of polyleucine and polyvaline atomistic references. Fig. 2 shows that (and this is theoretically guaranteed[4] for spline potentials) at a minimum of relative entropy, the atomistic and CG distribution functions for arguments of the different potential functions (e.g., bond lengths and angles, dihedral angles and pair distances) match quantitatively.

The only case where the all-atom and CG distributions do not match is the α carbon - sidechain or CS bond potential (2nd row, 3rd column in Fig. 2), in which the optimal CG bond-length distribution is a singly-peaked Gaussian while the all-atom reference shows a bimodal distribution. This is expected since in the extended ensemble, the atomistic distribution reflects a combination of the distinct leucine and valine bond lengths. By using a harmonic bond potential in the CG model, it is not possible to capture this bimodal character – nor would it be physically realistic since the Gō model strategy attempts to represent a universal, amino-acid-independent side chain site. Minimization of the relative entropy guides the optimal CS bond length parameter (and distribution) to an average between the two reference cases. Similarly, all interactions (bond, angle and nonbonded pair) involving side-chains admit several peaks in their corresponding distributions due to contribution from both leucine and valine polymers.

In supplementary files, we provide LAMMPS compatible input files for this forcefield. Optimized knot values for splined potentials (bond-angular, torsional, nonbonded pair, native interactions) are used to produce finely interpolated lookup tables that are contained in `leu_val.table` while `ff_leu_val.dat` provides callbacks to relevant tabulated data. The latter can be referenced in a LAMMPS input script after defining units, bounding box, and peptide topology. Note that this LAMMPS forcefield assumes `atomtypes` 1, 2 and 3 for N, C and O CG sites respectively. For polypeptides the sidechain would be `atomtype` 4, while for heterosequences with the Gō-like native and WCA non-native potentials, sidechains must be allotted separate `atomtypes`.

[1] J.W. Mullinax and W.G. Noid, The Journal of Chemical Physics, **131**, (2009).
[2] J.W. Mullinax and W.G. Noid, Proceedings of the National Academy of Sciences, **107**, (2010).
[3] R.S. Cahn, C.K. Ingold and V. Prelog, Angewandte Chemie, **5**, 385-415 (1966).
[4] A. Chaimovich and M.S. Shell, The Journal of Chemical Physics, **134**, (2011).

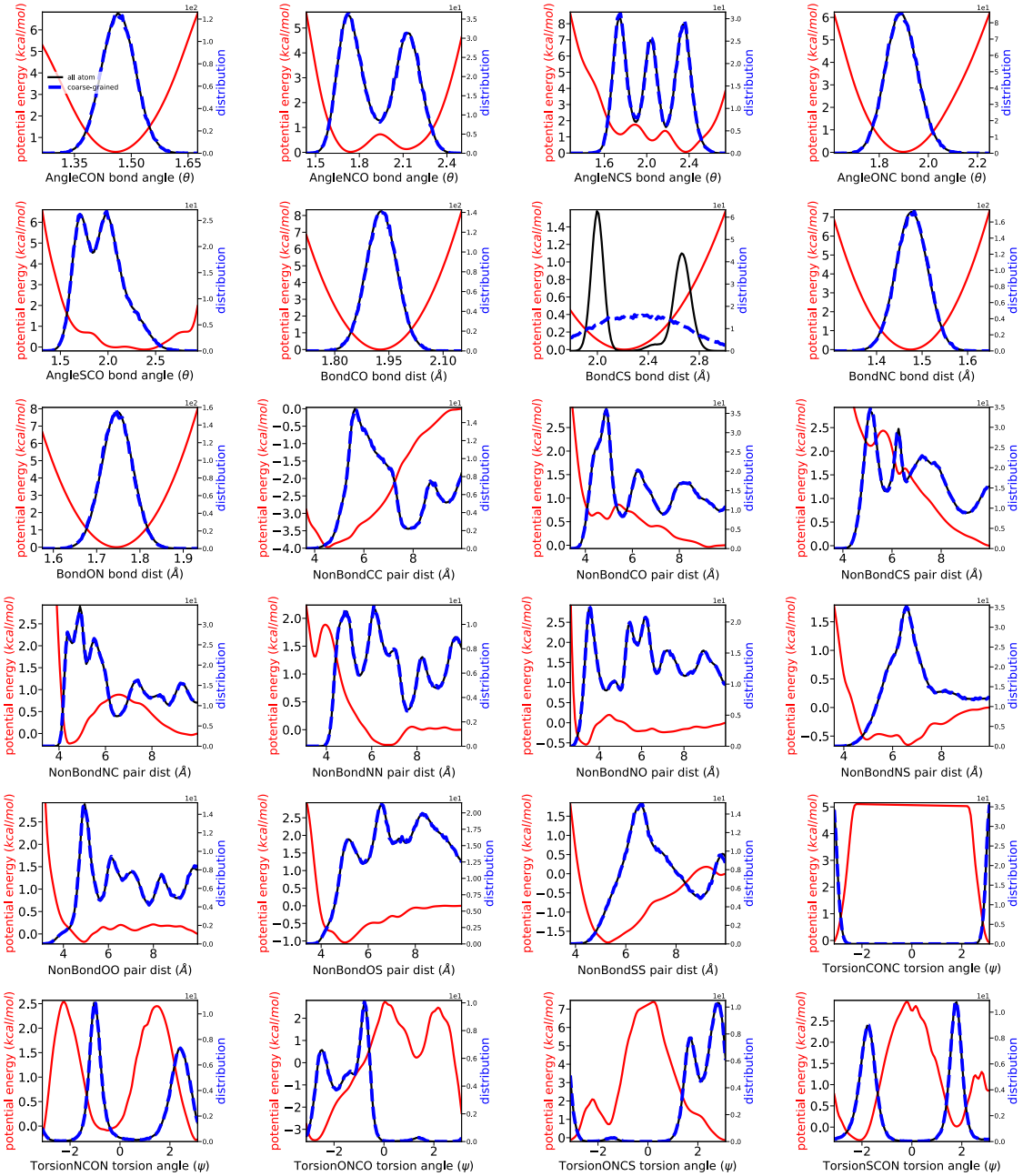

FIG. 2. Optimized interactions (intra-backbone, backbone-sidechain and inter-sidechain) for the $\mathrm{LEU}_{15} + \mathrm{VAL}_{15}$ extended-ensemble CG polypeptide model. Potential functions are in red, while black (atomistic) and blue (CG) lines show that at a relative entropy minimum, the atomistic and CG distribution functions for the argument of the potential function (bond lengths, bond angles, dihedral angles, pair distances) match quantitatively. Potentials are reported in kcal/mol, distances in Å, and angles in radians.